

MTH 204 : Lecture 05



Existence and Uniqueness of solution of IVP:

Ex: Consider the IVP : $|y'| + |y| = 0$, $y(0) = 1$

Note that $y \equiv 0$ is the only possible solution of the ODE $|y'| + |y| = 0$

But the IVP : $|y'| + |y| = 0$, $y(0) = 1$
has no solution.

Ex: Consider $y' = 2x$, $y(0) = 1$

Then $y = \int 2x dx + C \Rightarrow y = x^2 + C$

Now $y(0)=1 \Rightarrow C=1$

Hence $\boxed{y = x^2 + 1}$ is the unique solution of the above IVP.

Ex: Consider $xy' = y - 1$, $y(0) = 1$

Then $\frac{dy}{y-1} = \frac{dx}{x} \Rightarrow \ln(y-1) = \ln x + \ln C$

$$\Rightarrow y-1 = Cx \Rightarrow y = 1 + Cx$$

Now note that $y(0)=1$ for any C

Thus the above IVP has infinitely many solutions.

Existence Theorem:

Given the IVP: $y' = f(x, y)$, $y(x_0) = y_0$.

If $f(x, y)$ is continuous in a rectangle R
.....⁽¹⁾

$$R = \{ (x, y) \in \mathbb{R}^2 : |x - x_0| < a, |y - y_0| < b \}$$

and bounded in R (i.e. there exists $K > 0$
.....⁽²⁾

such that $|f(x, y)| \leq K \quad \forall (x, y) \in R$),

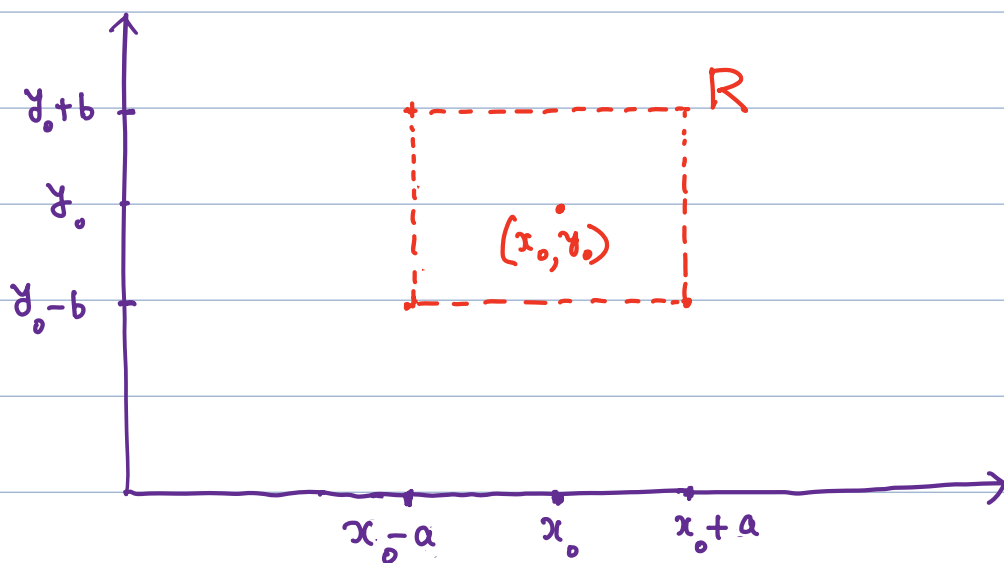
then the IVP has at least one solution $y(x)$.

This solution exists for all x in $|x - x_0| < \alpha$

where $\alpha = \min \left\{ a, \frac{b}{K} \right\}$.

i.e. the graph of the solution lies inside a Rectangle R_1 ($R_1 \subset R_2$)

where $R_1 = \{ (x, y) \in \mathbb{R}^2 : |x - x_0| < \min\{a, \frac{b}{k}\}, |y - y_0| < b \}$

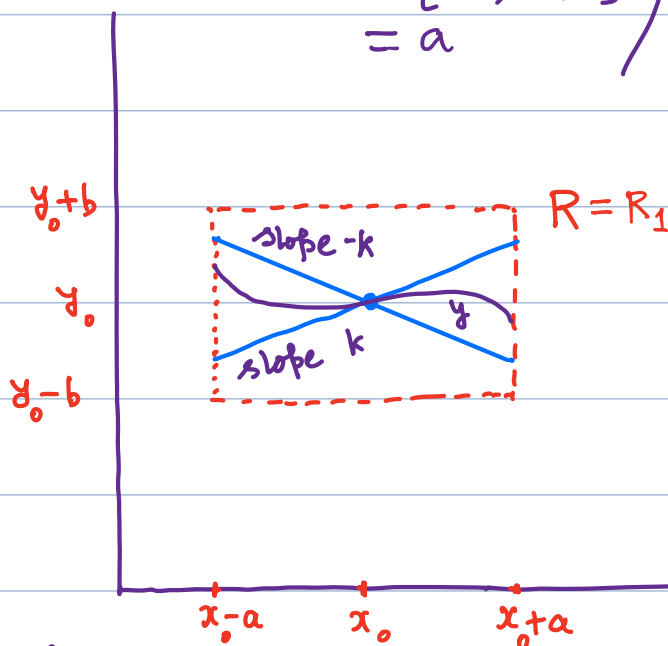


Note:

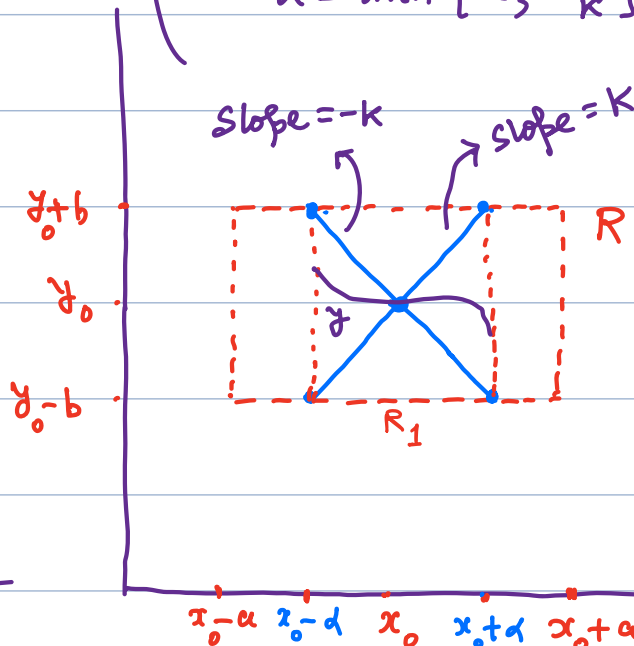
- $f(x, y)$ is bounded by K means
 $|y'| = |f(x, y)| \leq K$

i.e. The slope of any solution is at least $-K$ and at most K .

(In this case $\alpha = \min\{a, \frac{b}{k}\} = a$)



(In this case $\alpha = \min\{a, \frac{b}{k}\} = \frac{b}{k} < a$)



(The second condition of the existence theorem)

Uniqueness Theorem:

- Given the IVP : $y' = f(x, y)$, $y(x_0) = y_0$.
- Let the IVP meet the conditions of the Existence Theorem.

If $f_y = \frac{\partial f}{\partial y}$ is continuous in R and is ⁽³⁾
bounded in R (i.e. there exists $M > 0$ such ⁽⁴⁾
that $|f_y(x, y)| \leq M \quad \forall (x, y) \in R$), then the
IVP has a unique solution $y(x)$.

This solution exists for all x
in $|x - x_0| < d$ where $d = \min \left\{ a, \frac{b}{K} \right\}$.

i.e. the graph of the unique solution lies inside R_1

Note: (1) The above condition (4) can be replaced by a weaker condition "Lipschitz condition".

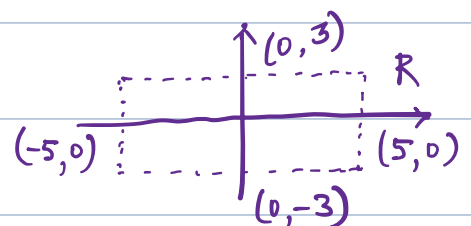
$$|f(x, y_2) - f(x, y_1)| \leq M |y_2 - y_1| \text{ whenever } (x, y_1), (x, y_2) \in R$$

However only continuity of $f(x, y)$ is not enough to guarantee the uniqueness of the solution.

(2) The proof of the above theorem requires some advanced topics of Analysis.

Ex (1): $y' = 1 + y^2$, $y(0) = 0$

$$f(x, y) = 1 + y^2, \quad x_0 = 0, \quad y_0 = 0$$



Let $R = \{ (x, y) \mid |x| < 5, |y| < 3 \}$

Then (1) f is continuous everywhere & so in R

(2) $|f(x, y)| \leq |1 + y^2| \leq 1 + |y|^2 < 1 + 3^2 = 10$

So, $f(x, y)$ is bounded in R with $K=10$

(3) $\frac{\partial f}{\partial y} = 2y$ is continuous everywhere and so in R

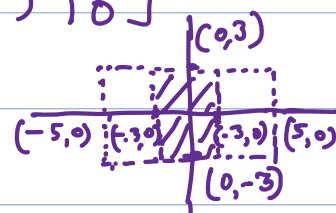
(4) $\left| \frac{\partial f}{\partial y} \right| \leq |2y| \leq 2|3| = 6$

So, $\frac{\partial f}{\partial y}$ is bounded in R with $M=6$

Hence a unique solution exists

for all x s.t. $|x| < \min\{5, \frac{3}{10}\}$

i.e. for $|x| < .3$



The graph of the solution lies in the rectangle $R_1 = \{ (x, y) : |x| < .3, |y| < 3 \}$

Ex ②: $|y'| + |y| = 0, y(0) = 1$

This IVP has no solution

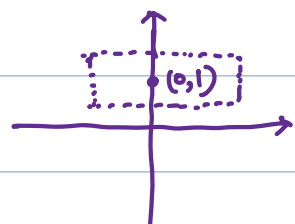
The only solution to $|y'| + |y| = 0$ is $y = 0$ but this doesn't satisfy the condition $y(0) = 1$

$$\begin{aligned} \text{Now } |y'| + |y| = 0 &\Rightarrow |y'| = -|y| \\ |y'| = -|y| &\Rightarrow \left. \begin{aligned} |y'| &= -y & \text{if } y \geq 0 \\ &y & \text{if } y < 0 \end{aligned} \right\} \end{aligned}$$

$$\Rightarrow y' = \left. \begin{aligned} -y & \text{ if } y \geq 0, y' \geq 0 \\ y & \text{ if } y < 0, y' \geq 0 \\ y & \text{ if } y \geq 0, y' < 0 \\ -y & \text{ if } y < 0, y' < 0 \end{aligned} \right\} = \boxed{f(x, y)}$$

$f(x, y)$ is not defined at any point other than $(0, 0)$ and so the conditions (1) and (2) are not satisfied. Hence, the existence theorem can't be applied here.

Ex (3): $xy' = y - 1$ $y(0) = 1$



$$\Rightarrow y' = \frac{y-1}{x} = f(x, y)$$

$$\Rightarrow \frac{dy}{y-1} = \frac{dx}{x} \Rightarrow \ln|y-1| = \ln|x| + C, \Rightarrow \boxed{y = 1 + Cx}$$

So, there are infinitely many solutions.

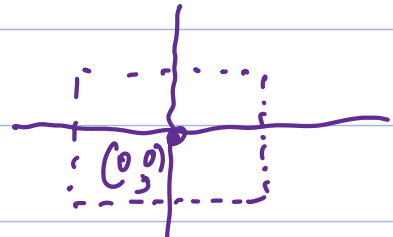
$f(x, y)$ is not defined on any point on the y -axis and hence in any rectangle centered at $(0, 1)$ (it will contain points from y -axis)

So, the existence theorem can't be applied here.

However the IVP has infinitely many solutions

$$y(x) = 1 + cx \text{ for any constant } c$$

Ex (7): $y' = \sqrt{|y|}$, $y(0) = 0$



$f(x, y) = \sqrt{|y|}$ continuous everywhere }
and bounded in any rectangle }
centered at $(0, 0)$

So, by existence theorem, a solution exists.

In fact $y(x) = \begin{cases} x^2/4 & x \geq 0 \\ -x^2/4 & x < 0 \end{cases}$

and $y \equiv 0$ are both solutions.

Note that $\frac{\partial f}{\partial y}$ does not exist at $y=0$

The Lipschitz condition is also violated at any region that includes 0.

We take $y_1=0$, $y_2>0$. Then
$$\frac{f(x, y_2) - f(x, y_1)}{|y_2 - y_1|} = \frac{\sqrt{|y_2|} - 0}{|y_2|} = \frac{\sqrt{y_2}}{y_2} = \frac{1}{\sqrt{y_2}} \rightarrow \infty \text{ as } y_2 \rightarrow 0$$

Thus Uniqueness Theorem can't be applied here.

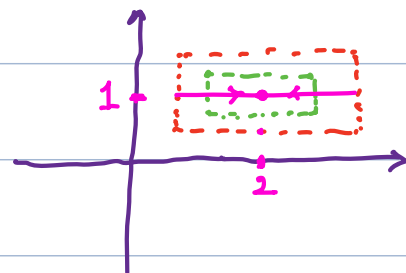
Here we are getting two solutions of the IVP.

Ex: $(x-2)y' = y$, $y(2) = 1$

The above IVP can be written as $y' = \frac{y}{x-2}$

So, $f(x, y) = \frac{y}{x-2}$ $(x_0, y_0) = (2, 1)$

Now $f(x, 1) = \frac{1}{x-2}$



$$\lim_{x \rightarrow 2^+} f(x, 1) = \infty, \quad \lim_{x \rightarrow 2^-} f(x, 1) = -\infty$$

$\Rightarrow f(x, y)$ is unbounded in any rectangle centered at $(2, 1)$

$\Rightarrow$ The existence theorem can't be applied here.

Actually from $(x-2)y' = y$

$$\text{we get } \frac{dy}{y} = \frac{dx}{x-2} \Rightarrow \ln|y| = \ln|x-2| + C,$$

$$\Rightarrow y = c(x-2)$$

But when $x=2$, $y=0$

Thus the initial condition $y(2)=1$ is not satisfied.

Hence this IVP has no solution.

Ex: $y' = x+y$, $y(0)=3$

Here $f(x,y) = x+y$

- $f(x,y)$ is continuous and bounded in any rectangle R centered at $(0,3)$ $R = \{(x,y) : |x-0| < a, |y-3| < b\}$
- $\frac{\partial f}{\partial y} = 1$ is continuous and bounded in R

Hence by the existence and uniqueness theorem,

the IVP has a unique solution $y(x)$ for

all x such that $|x-0| < \alpha$

where $\alpha = \min\{a, \frac{b}{k}\}$ where $|f(x,y)| \leq k$

$$\forall (x,y) \in R$$

Here $|f(x,y)| = |x+y| \leq |x| + |y| \leq a + 3 + b = a + b + 3$
 $\forall x, y \in R$

So, k can be taken as $a + b + 3$

$$\text{So, } \alpha = \min \left\{ a, \frac{b}{a+b+3} \right\}$$

and the graph of the solution lies

$$\text{in } R' = \left\{ (x, y) \in \mathbb{R}^2 : |x-0| < \alpha, |y-3| < b \right\} \subset \mathbb{R}$$

Note: $y' = x + y \Rightarrow \frac{dy}{dx} - y = x$

$$\text{I.F.} = e^{-\int 1 dx} = e^{-x}$$

Multiplying both sides by e^{-x} we get

$$(y e^{-x})' = \int x e^{-x} dx + C$$

$$y e^{-x} = -x e^{-x} - e^{-x} + C$$

$$\Rightarrow y = -x - 1 + C e^x$$

$$\text{Now } y(0) = 3 \Rightarrow 3 = -0 - 1 + C \Rightarrow C = 4$$

Thus the unique solution of the IVP is

$$\boxed{y = -x - 1 + 4e^x}$$

Ex: ① Does the initial value problem

$$(x-1)y' = 2y, \quad y(1) = 1 \text{ have a solution?}$$

Does your result contradict the existence theorem?

② What happens if you replace $y(1)=1$ by $y(1)=k$

$$\textcircled{1} \quad (x-1)y' = 2y \Rightarrow \frac{dy}{y} = \frac{2 dx}{(x-1)} \\ \Rightarrow \ln|y| = \ln|x-1|^2 + C_1 \Rightarrow \boxed{y = C(x-1)^2}$$

This is the general solution.

Note that $y(1)=0$ for any value of C .

Therefore $y(1)=1$ can't be satisfied.

However this doesnot contradict the theorem.

because the differential equation can be rewritten

$$\text{as : } y' = \frac{2y}{x-1} \quad \text{and} \quad f(x,y) = \frac{2y}{x-1}$$

is not defined when $x=1$

② For $k \neq 0$, we will get no solution.

Existence of the solution of IVP is not satisfied.

For $k=0$, the initial condition becomes

$$y(1)=0 \quad \text{and the solution } y = C(x-1)^2 \\ \text{satisfies } y(1)=0 \text{ for any } C.$$

Thus we obtain infinitely many solution.

Thus uniqueness of the solution of IVP is not satisfied but it doesnot contradict the

theorem since $f(x) = \frac{2y}{x-1}$ is not defined when $x=1$.

③ Find all initial conditions such that $(x^2 - 4x)y' = (2x - 4)y$ has no solution, precisely one solution and more than one solution.

$$(x^2 - 4x)y' = (2x - 4)y$$
$$\Rightarrow \frac{dy}{y} = \frac{(2x - 4)}{(x^2 - 4x)} dx \Rightarrow \ln|y| = \ln|x^2 - 4x| + C_1$$
$$\Rightarrow \boxed{y = c(x^2 - 4x)}$$

Thus we can see that the above has

(1) no solution if $y(0) = K \neq 0$ or $y(4) = K \neq 0$ }

(2) Infinitely many solution if $y(0) = 0$ or $y(4) = 0$
(It will be a solution for any value of c)

(3) Unique solution if $y(x_0) = y_0$ for any y_0
and $x_0 \neq 0$ or $x_0 \neq 4$

This does not contradict the theorem because

$$f(x, y) = \frac{2x - 4}{x^2 - 4x} \text{ is not defined when } x = 0 \text{ or } x = 4.$$

Second Order ODEs:

Second order Linear ODE is of the form:

$$y'' + p(x)y' + q(x)y = r(x)$$

It is homogeneous if $r(x) = 0$

Ex: • $y'' + 25y = e^{-x} \cos x$: Linear, Non-homogeneous

• $xy'' + y' + xy = 0$

$\Rightarrow y'' + \frac{1}{x}y' + y = 0$: Linear, homogeneous

• $y''y + (y')^2 = 0$: Non-linear

Principle of Superposition:

The linear combination of any two solutions of a second order linear homogeneous ODE is again a solution. That is, if y_1 and y_2 are solutions then $c_1y_1 + c_2y_2$ is also a solution for any constants c_1 and c_2 .

Proof: Given: $y_1'' + p(x)y_1' + q(x)y_1 = 0$
and $y_2'' + p(x)y_2' + q(x)y_2 = 0$

$$\text{Then } (c_1 y_1 + c_2 y_2)'' + p(x)(c_1 y_1 + c_2 y_2)' + q(x)(c_1 y_1 + c_2 y_2)$$

$$= c_1 y_1'' + c_2 y_2'' + p(x)(c_1 y_1' + c_2 y_2') + q(x)(c_1 y_1 + c_2 y_2)$$

$$= c_1 (y_1'' + p(x)y_1' + q(x)y_1) + c_2 (y_2'' + p(x)y_2' + q(x)y_2)$$

$$= c_1 \cdot 0 + c_2 \cdot 0 = 0$$

Ex: $y'' + y = 0$

$y_1 = \cos x$, $y_2 = \sin x$ are solutions $y_1' = -\sin x$, $y_2' = \cos x$

Then $y = c_1 \cos x + c_2 \sin x$

is also a solution.

$y_1'' = -\cos x$, $y_2'' = -\sin x$
 $y_1'' + y_1 = 0$, $y_2'' + y_2 = 0$

Ex: The above doesn't work with non-linear ODEs.

e.g: $y''y - xy' = 0$

$y_1 = x^2$, $y_2 = 1$

$y_2' = 0$, $y_2'' = 0$

$y_1'' = 2$, $y_1' = 2x$

$0 \cdot y - x \cdot 0 = 0$

$y_1'' y_1 - x y_1' = 2x^2 - x \cdot 2x = 2x^2 - 2x^2 = 0$

Let $y = y_1 + y_2 = 1 + x^2$

$y' = 2x$, $y'' = 2$

Then $y''y - xy' = 2(1+x^2) - x(2x) = 2 + 2x^2 - 2x^2$

$\Rightarrow y''y - xy' = 2 \neq 0$

Ex: The above doesn't work with nonhomogeneous ODE:

e.g. $y'' + y = 1$

$$y_1 = 1 + \cos x, \quad y_2 = 1 + \sin x$$

$$y_1' = -\sin x$$

$$y_1'' = -\cos x$$

$$y_2' = \cos x, \quad y_2'' = -\sin x$$

$$y_1'' + y_1 = -\cos x + 1 + \cos x = 1$$

$$y_2'' + y_2 = -\sin x + 1 + \sin x = \underline{1}$$

$$y = y_1 + y_2 = 2 + \cos x + \sin x$$

$$y' = -\sin x + \cos x, \quad y'' = -\cos x - \sin x$$

$$\text{Now } y'' + y = -\cos x - \sin x + 2 + \cos x + \sin x = 2 \neq 1$$

So, $y = y_1 + y_2$ is not a solution.

